MACHINE LEARNING -1 - PROJECT - GUIDED

STATE WISE HEALTH INCOME & HAIR SALON

2/10/2024

Data Mining

Part 1: Clustering

Context

The State_wise_Health_income.csv dataset given is about the Health and economic conditions in different States of a country. The Group States based on how similar their situation is, so as to provide these groups to the government so that appropriate measures can be taken to escalate their Health and Economic conditions.

Objective

The objective of the analysis is to cluster the states based on the given dataset "State_wise_Health_income.csv," considering both health and economic indicators. By grouping states with similar conditions, the goal is to provide the government with valuable insights for implementing targeted measures to improve the health and economic conditions of each group.

Data Description

1. **States**: Names of States
2. **Health_indeces1**: A composite index rolls several related measures (indicators) into a single score that provides a summary of how the health system is performing in the State.
3. **Health_indeces2**: A composite index rolls several related measures (indicators) into a single score that provides a summary of how the health system is performing in certain areas of the States.
4. **Per_capita_income**: Per capita income (PCI) measures the average income earned per person in a given area (city, region, country, etc.) in a specified year. It is calculated by dividing the area's total income by its total population.
5. **GDP**: GDP provides an economic snapshot of a country/state, used to estimate the size of an economy and growth rate.

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Data Analysis & A detailed solution around the problems:

Overview of the dataset: The initial steps to get an overview of any dataset are to:

- Observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not.
- Get information about the number of rows and columns in the dataset.
- Find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- Check the statistical summary of the dataset to get an overview of the numerical columns of the data.

Observation 1: The dataset has 296 rows and 6 columns.

Observation 2: All the columns in the data are non-null. Only States column is categorical (object) type rest are numerical (integer) hence it would be easy to analyze the dataset.

Observation 3: It's very crucial to identify the missing values, and treat them before moving ahead with the data analysis, and as checked there are **no missing values** in the data, and we good to proceed.

Observation 4: There are 296 unique States in the data. And a **column 'unnamed: 0'** with no important data needs to be dropped for further analysis.

Observation 5: Statistical summary is nothing but the technique to identify variables such as min, max, mean, standard deviation, count, 25th, 50th, & 75th percentile of the data.

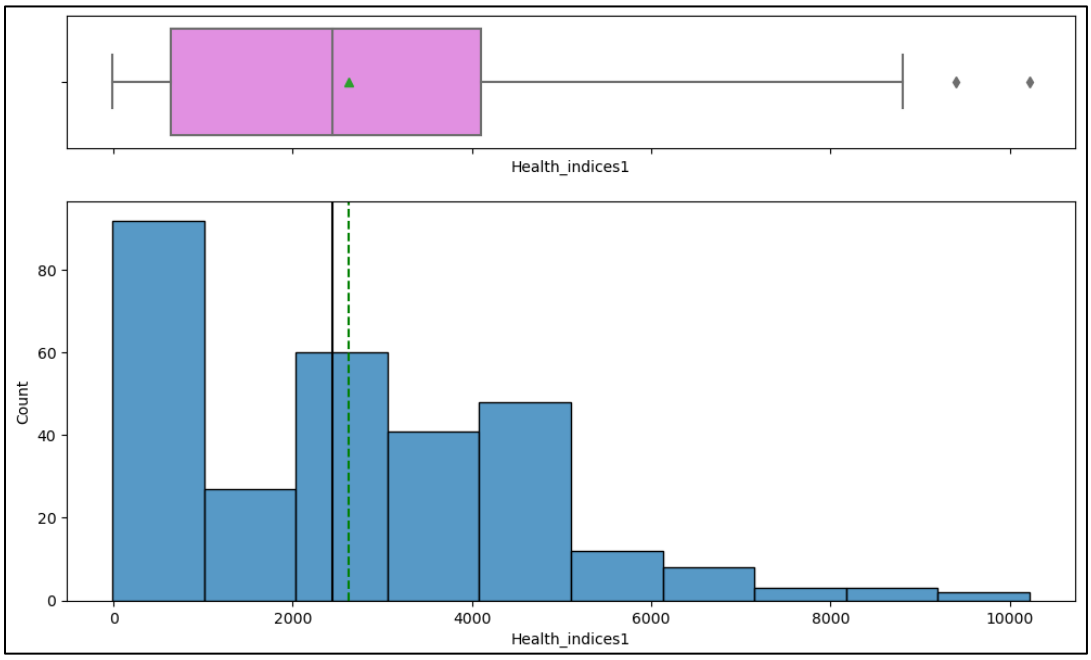
	count	mean	std	min	25%	50%	75%	max
Health_indices1	296.0	2629.195946	2041.890970	-10.0	640.0	2446.5	4102.75	10219.0
Health_indices2	296.0	693.594595	469.738035	0.0	173.5	810.5	1076.00	1508.0
Per_capita_income	296.0	2159.597973	1493.663013	500.0	746.0	1869.0	3138.75	7049.0
GDP	296.0	174717.050676	167439.128587	22.0	8679.5	135748.5	314751.25	728575.0

- Per capita income min. is 500 and max is 7049, with an average of 2159.
- GDP min is 22 and max is 728575, with an average of 174717.
- Health Indices 1 min is -10 and max is 10219, with mean of 2629.
- Health Indices 2 min is 0 and max is 1508, with mean of 693.

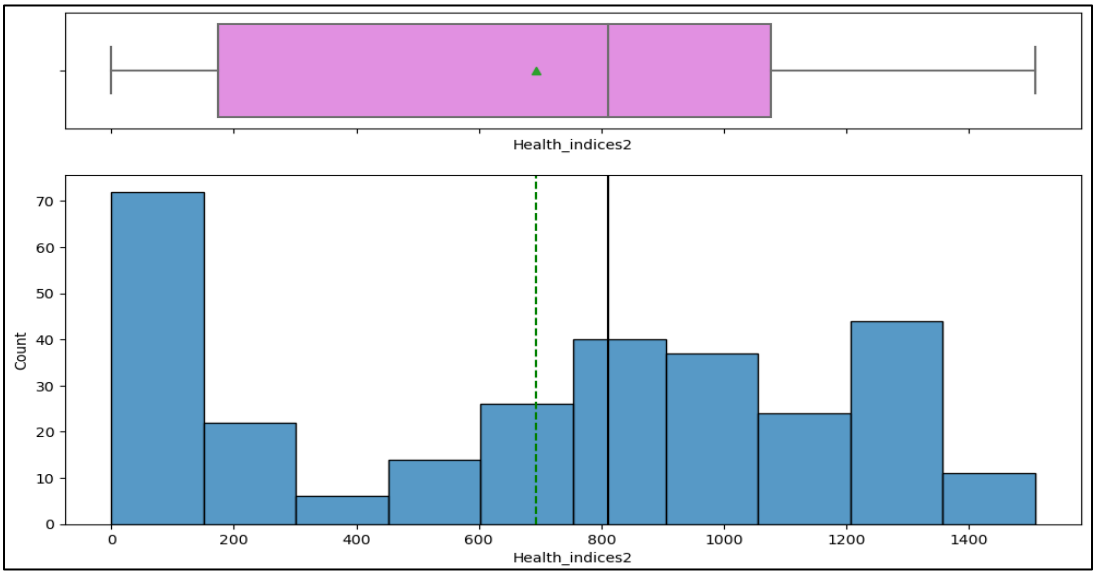
Topic 1: Exploratory Data Analysis:

1. Univariate analysis: (The green dotted lines in all the image denotes average of the data)

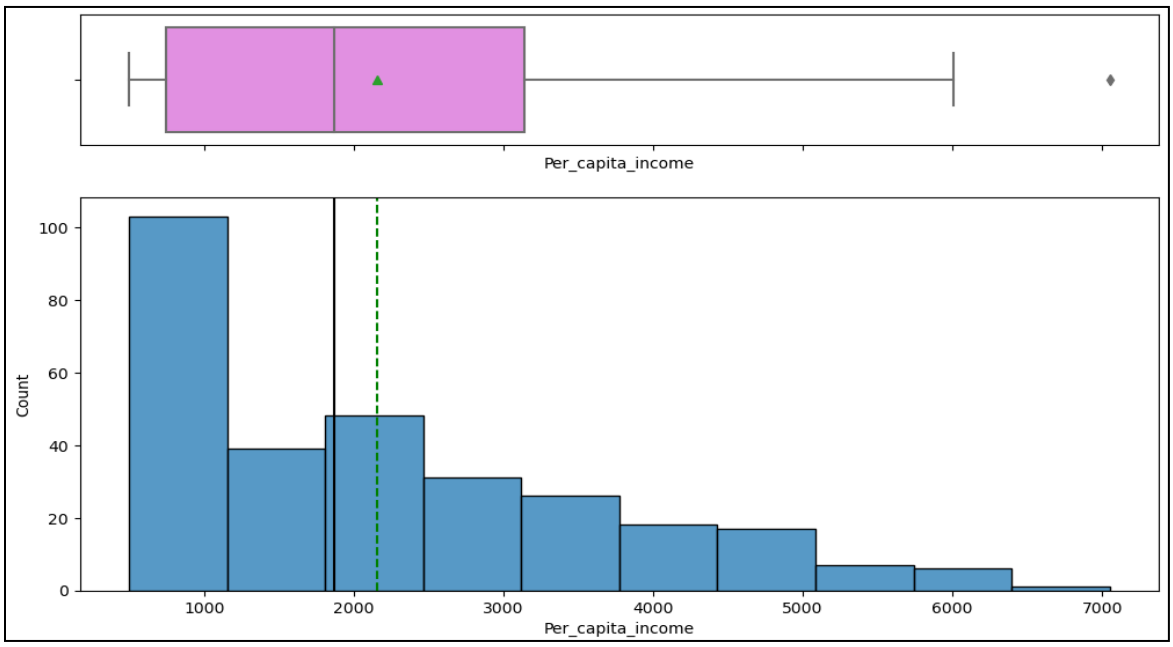
1.1 Health Indices 1: Outliers observed in Health Indices 1, which needs to be treated for further analysis. The count mainly ranges from 0 to 5000 and slows down eventually after that, with highest from 0 to 1000.



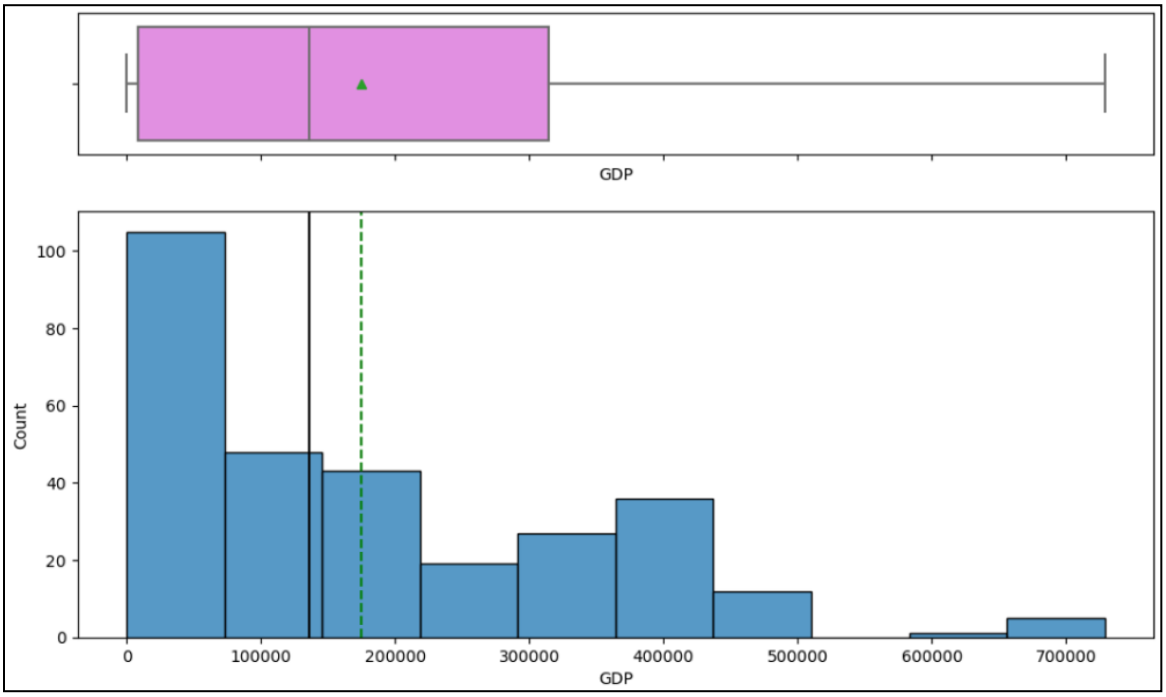
1.2 Health Indices 2: No outliers found in the Health Indices 2. Data spread from count 0 to 1500.



1.3 Per Capita Income: Outlier found in this segment. Data count starting from 500 and diminishing towards 7000. Highest range between 500 to 1100.



1.4 GDP: No outlier found in the GDP. Similar pattern found like per capita income, data starting from 0 and narrowing towards 700000. Maximum count find below 100000.

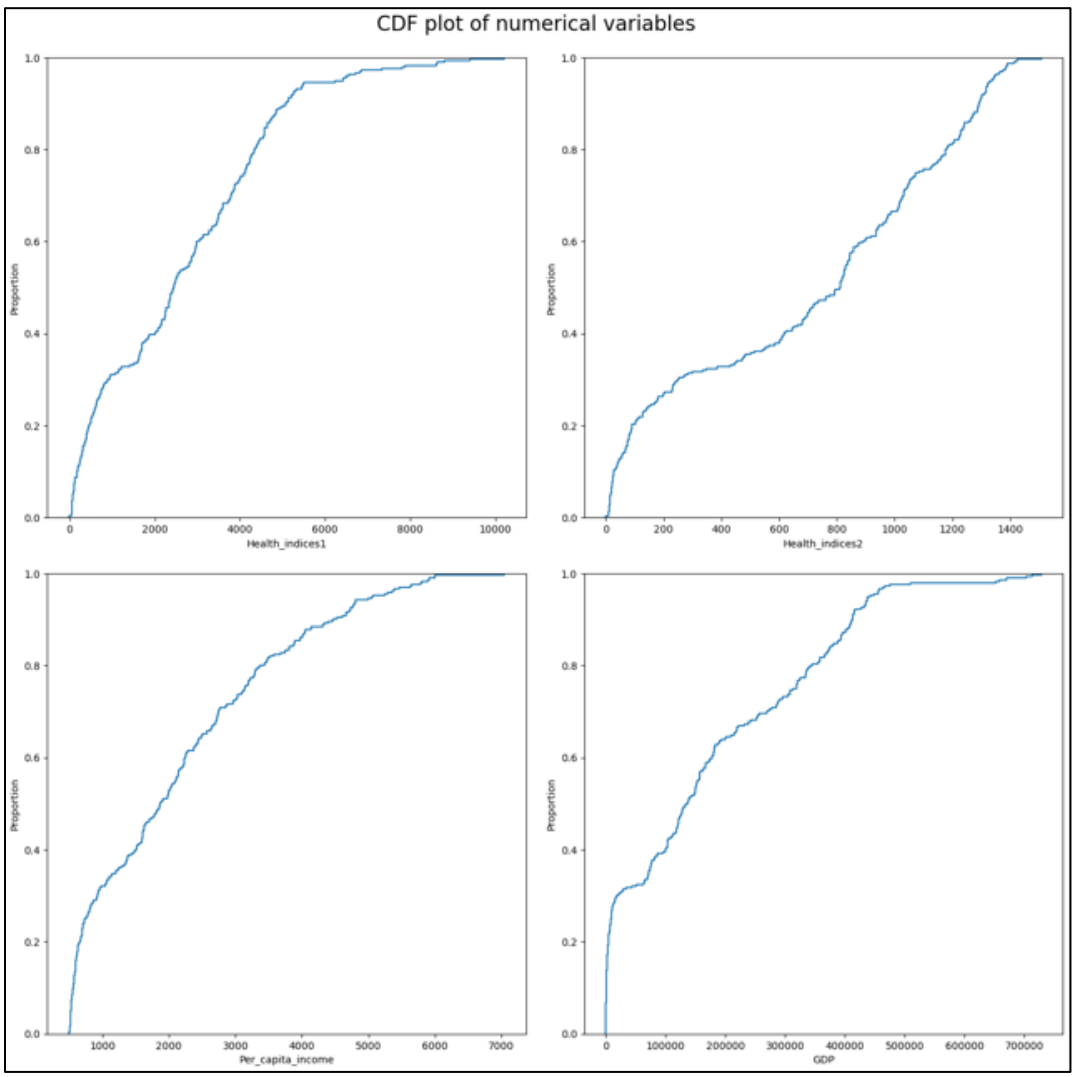


1.5 States: We have 296 unique states, and for data analysis categorical column contribute the least.

CDF Plot: CDF plot is a versatile tool in statistics and data analysis, offering a comprehensive view of the distribution of a random variable and aiding in various statistical tasks such as comparison, quantile estimation, hypothesis testing, and model evaluation. **Skewness for each variable is as below:**

```
GDP                  0.826397
Per_capita_income    0.818014
Health_indices1      0.715652
Health_indices2     -0.173267
```

- Health indices 2 is negatively skewed and also right skewed except for others.



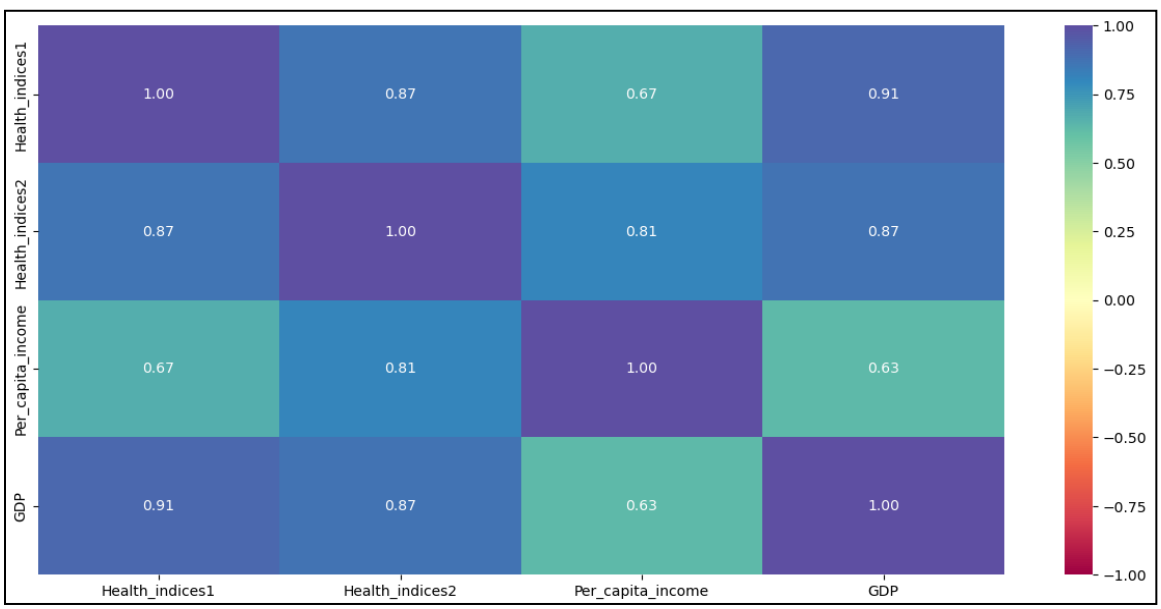
2. Bivariate Analysis: Let's check for correlations:

2.1 Heatmap: Provides a visual and intuitive way to explore and interpret complex datasets, making them a valuable tool in various fields for tasks such as correlation analysis, pattern recognition, and data exploration.

Observations:

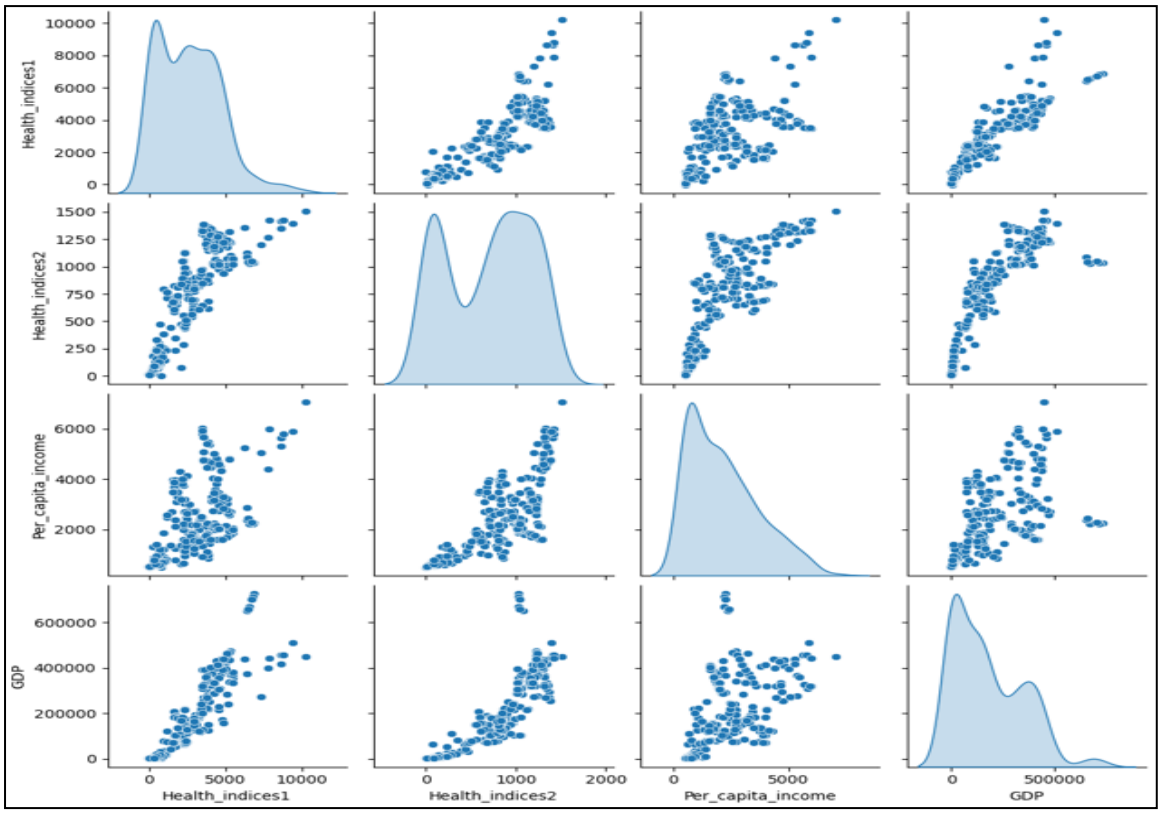
- Multi-collinearity observed in the data.
- Health Indices 1 is highly correlated with GDP. Health Indices 2 and GDP are also decently correlated.
- Health Indices 1 and Per capita income show a moderate correlation as compared to Health indices 2 and Per capita income with slightly high correlation.
- Again Per capita income and GDP show a moderate correlation.

		correlation
Health_indices1	GDP	0.907190
Health_indices2	GDP	0.869464
Health_indices1	Health_indices2	0.866421
Health_indices2	Per_capita_income	0.811987
Health_indices1	Per_capita_income	0.669226
Per_capita_income	GDP	0.629372



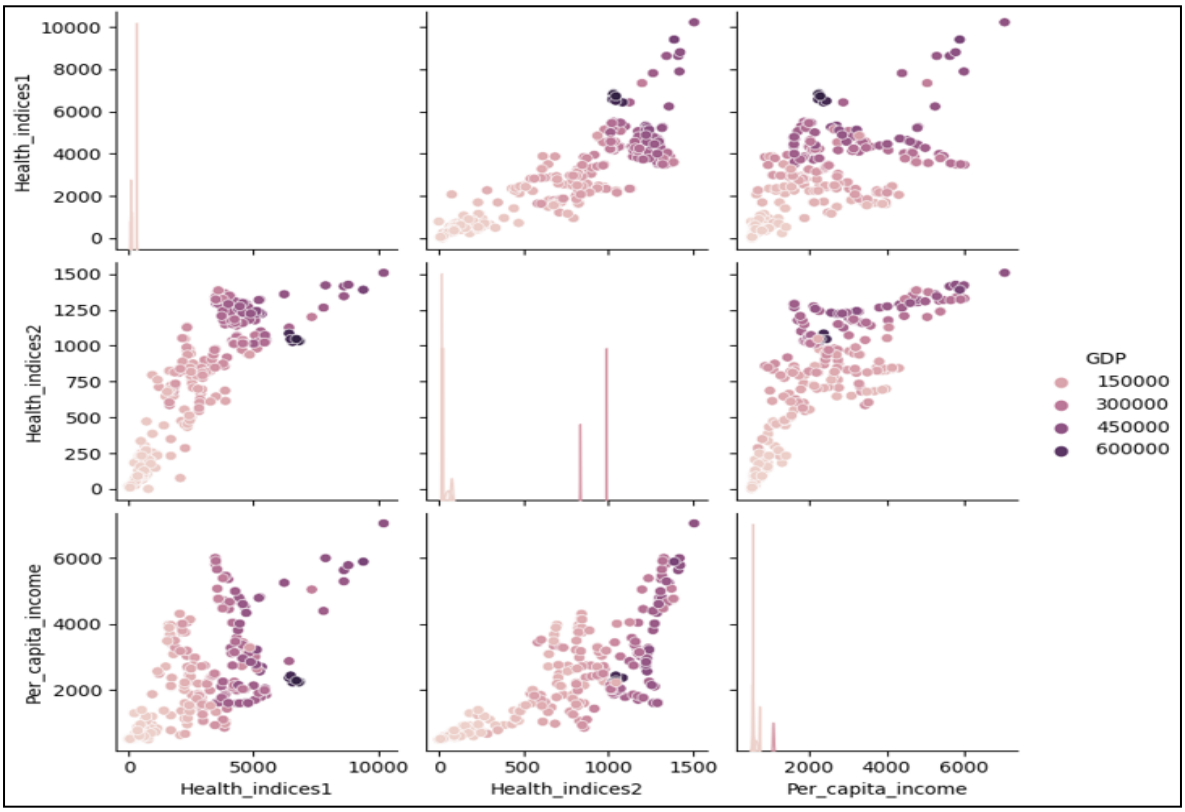
2.2 Pairplots: Provide a comprehensive and visual approach to exploring relationships between multiple variables in a dataset. They are especially useful when dealing with datasets with several features, enabling analysts to gain insights into the overall structure and characteristics of the data.

	Health_indices1	Health_indices2	Per_capita_income	GDP
Health_indices1	1.000000	0.866421	0.669226	0.907190
Health_indices2	0.866421	1.000000	0.811987	0.869464
Per_capita_income	0.669226	0.811987	1.000000	0.629372
GDP	0.907190	0.869464	0.629372	1.000000



- **We can add a hue and see if we can see some clustered distributions:**

		correlation
Health_indices1	GDP	0.907190
Health_indices2	GDP	0.869464
Health_indices1	Health_indices2	0.866421
Health_indices2	Per_capita_income	0.811987
Health_indices1	Per_capita_income	0.669226
Per_capita_income	GDP	0.629372

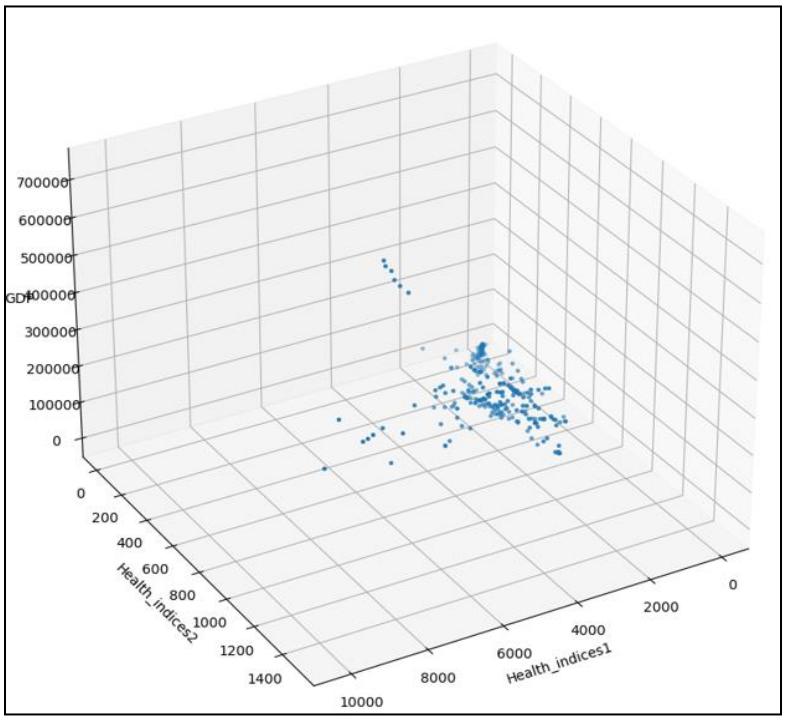


Let's visualize the modes of contacting the bank in a 3D plot:

A 3D plot is a graphical representation of three-dimensional data, allowing for the visualization of relationships among three variables. In a 3D plot, three axes are used to represent the three variables, and points or surfaces in the plot correspond to specific combinations of values for those variables.

Observations:

- Here we are using three parameters namely: Health Indices 1, Health Indices 2 & GDP.
- We see a cluster of data in between 0 to 2000 for health indices 1, between 0 to 800 for health indices 2 and maximum under 100000 for GDP, with few scatter from 100000 to 300000.



Topic 2: Data Preprocessing:

Outlier Detection -

- Let's find outliers in the data using z-score using the threshold you got in the above plot.

```
The following are the outliers in the data:

 Health_indices1 : [8802, 9403, 10219]

 Health_indices2 : []

 Per_capita_income : [7049]

 GDP : [703190, 713295, 728575]
```

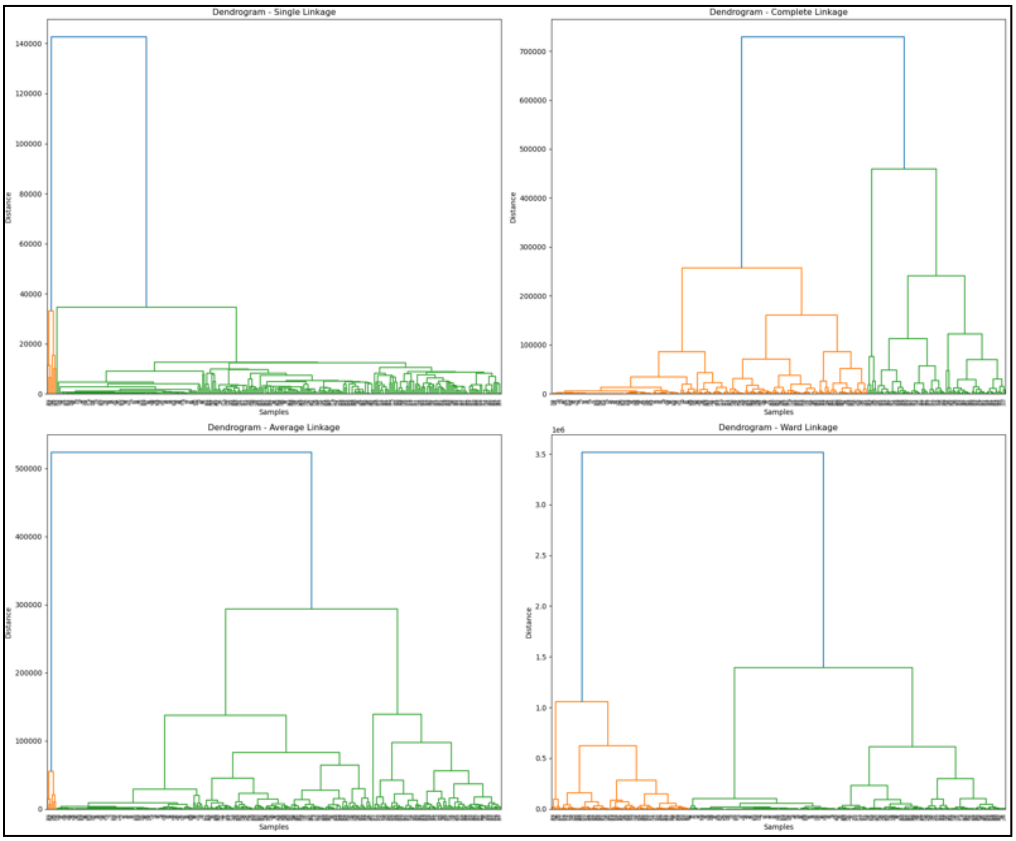
- We can treat outliers with min-max substitution if needed.

Scaling - Let's scale the data before we proceed with clustering.

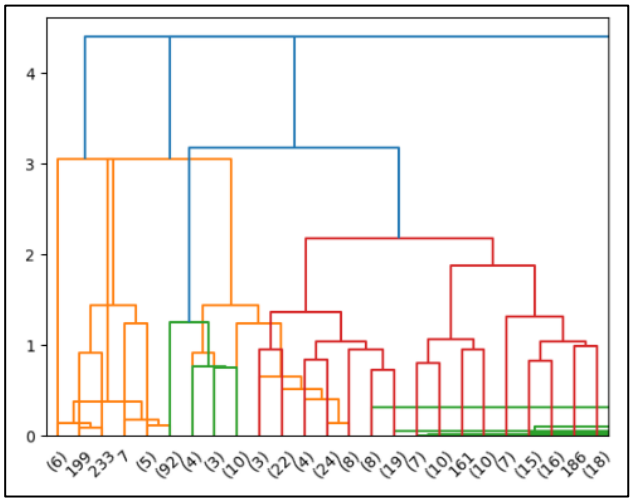
- Scaling the data before applying clustering algorithms helps to ensure that the algorithms perform optimally, producing meaningful and unbiased clusters based on the inherent patterns in the data rather than being influenced by variable magnitudes or units.

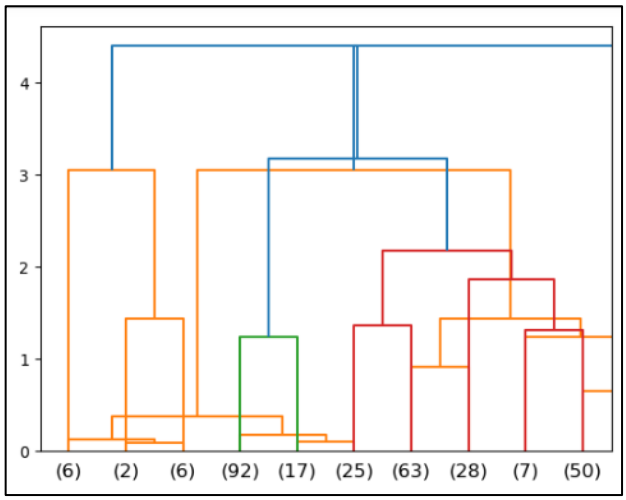
	Health_indices1	Health_indices2	Per_capita_income	GDP
0	-1.085240	-1.338315	-1.070054	-1.034327
1	-0.561310	-0.101493	0.369115	-0.604555
2	-0.968975	-0.841454	-0.707914	-0.881805
3	-1.195619	-1.425745	-1.064018	-1.043738
4	-1.268714	-1.461997	-1.094197	-1.045102

<u>Topic 3: Hierarchical Clustering</u>: Calculating and presenting linkage using different methods:



- Now here we use average linkage method and cut the dendograms with suitable clusters for better visualization (P=10, P=25).





Observation:

- Number of clusters determined using Ward linkage is 292.
- It is clear from the above dendrogram that 3 clusters should be formed.
- And hence we create three clusters and store it under clusters-3.
- The three group cluster gives a pattern based on high, medium, and low GDP.

	Health_indices1	Health_indices2	Per_capita_income	GDP	clusters-3
0	417	66	564	1823	2
1	1485	646	2710	73662	3
2	654	299	1104	27318	2
3	192	25	573	250	2
4	43	8	528	22	2

Silhouette Score:

For n_clusters = 2, silhouette score is 0.5362182000947584
For n_clusters = 3, silhouette score is 0.5214876308115696
For n_clusters = 4, silhouette score is 0.5059966784091774
For n_clusters = 5, silhouette score is 0.5326070186102602
For n_clusters = 6, silhouette score is 0.5253180820121315
For n_clusters = 7, silhouette score is 0.5477771423586254
For n_clusters = 8, silhouette score is 0.527926250016339
For n_clusters = 9, silhouette score is 0.5154033217352417

Creating final model:

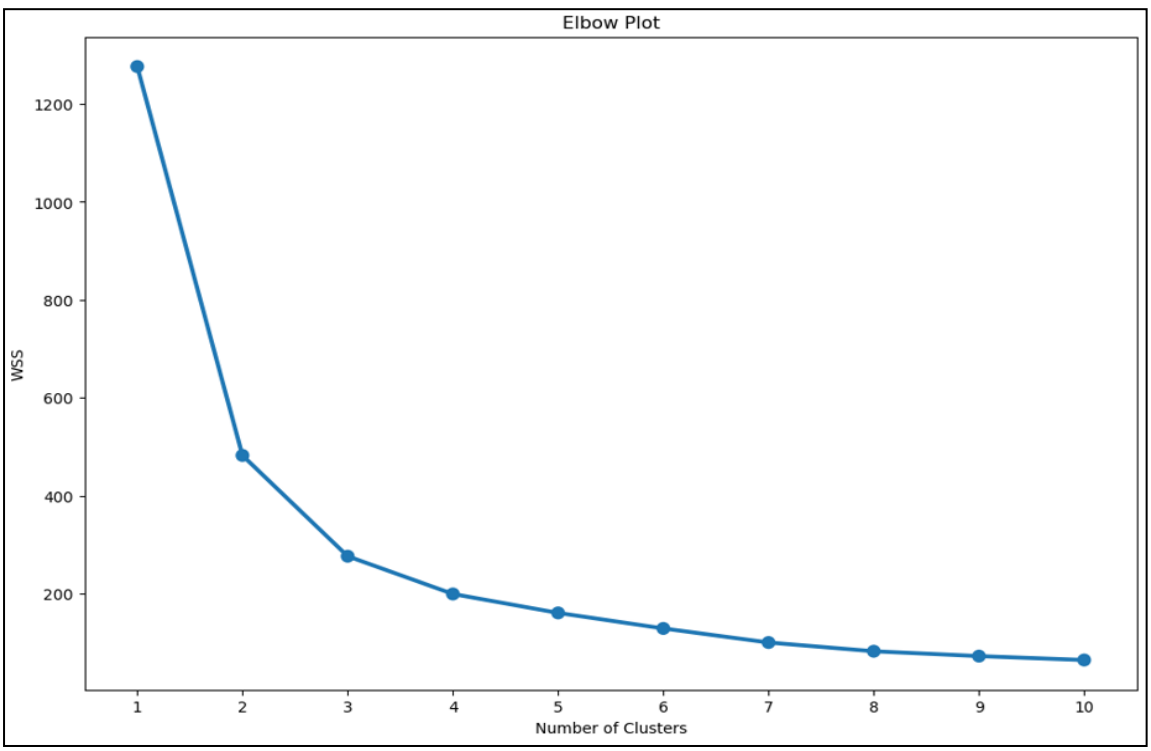
CPU times: total: 15.6 ms
Wall time: 11.7 ms

Topic 4: K-means Clustering: Is widely used in large dataset applications, the 'means' in K-means refers to averaging of the data; that is finding the centroid. The within cluster sum of squares is as follows:

```
[1276.5101351351352,
 482.1559255714835,
 276.2561392867222,
 199.1581672071259,
 160.47211144131273,
 128.76075478370933,
 100.00058012667091,
 82.15003059375553,
 72.23197345325109,
 64.25547258255443]
```

Checking Elbow Plot: Let's look for the point on the plot where there is a clear change in the slope, forming an elbow-like shape.

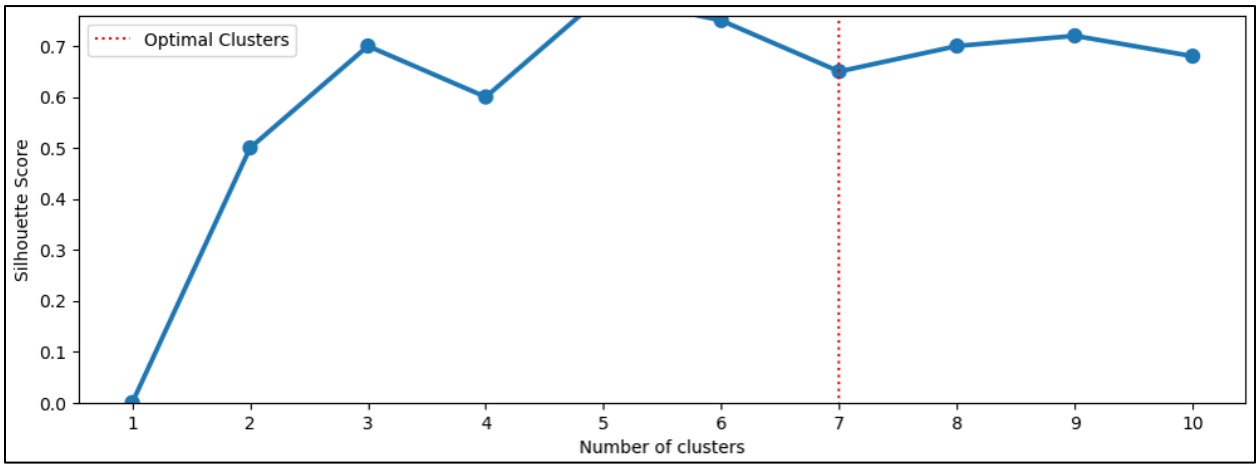
The elbow point is typically the optimal number of clusters. And here it can be either k=3 or 4.



Checking Silhouette Scores:
The Average Silhouette Score for 2 clusters is 0.5671
The Average Silhouette Score for 3 clusters is 0.53477
The Average Silhouette Score for 4 clusters is 0.55527
The Average Silhouette Score for 5 clusters is 0.56204
The Average Silhouette Score for 6 clusters is 0.58121
The Average Silhouette Score for 7 clusters is 0.55049
The Average Silhouette Score for 8 clusters is 0.53255
The Average Silhouette Score for 9 clusters is 0.51682
The Average Silhouette Score for 10 clusters is 0.49929

From the silhouette scores, find which k value has highest value and have elbow in elbow curve.

- Silhouette width ranges between -1 to +1 and a value 1 is a good cluster.



	Health_indices1	Health_indices2	Per_capita_income	GDP	HC_Clusters	K_means_segments
0	417	66	564	1823	2	0
1	1485	646	2710	73662	1	3
2	654	299	1104	27318	2	0
3	192	25	573	250	2	0
4	43	8	528	22	2	0

Cluster Profiling is given below: It is a DataFrame that contains the mean values of features for each K-means cluster. Each row corresponds to a cluster, and each column represents a feature.

```
0    107
3     90
5     55
2     30
4      8
1      6
```

K_means_segments	Health_indices1	Health_indices2	Per_capita_income	GDP	HC_Clusters	freq
0	604.44	137.00	732.34	12979.14	2.00	107
1	6649.33	1044.00	2299.83	687649.67	0.00	6
2	4041.27	1302.30	4823.93	341640.23	1.00	30
3	2577.44	792.48	2521.77	142134.09	1.02	90
4	8589.38	1371.00	5635.25	425379.62	0.00	8
5	4577.24	1145.84	2369.51	359223.07	1.00	55

Cluster Profiling and Comparison:

- **Cluster Profiling: K-means Clustering** – The clusters here have been formed and assigned with the respective count in each segment.

K_means_segments	Health_indices1	Health_indices2	Per_capita_income	GDP	HC_Clusters	count_in_each_segment
0	604.439252	137.000000	732.336449	12979.140187	2.000000	107
1	6649.333333	1044.000000	2299.833333	687649.666667	0.000000	6
2	4041.266667	1302.300000	4823.933333	341640.233333	1.000000	30
3	2577.444444	792.477778	2521.766667	142134.088889	1.022222	90
4	8589.375000	1371.000000	5635.250000	425379.625000	0.000000	8
5	4577.236364	1145.836364	2369.509091	359223.072727	1.000000	55

- **Cluster Profiling: Hierarchical Clustering** – It is another clustering algorithm that organizes data points into a tree-like structure, known as a dendrogram. This type of clustering does not require specifying the number of clusters beforehand, and it can be agglomerative (bottom-up) or divisive (top-down).

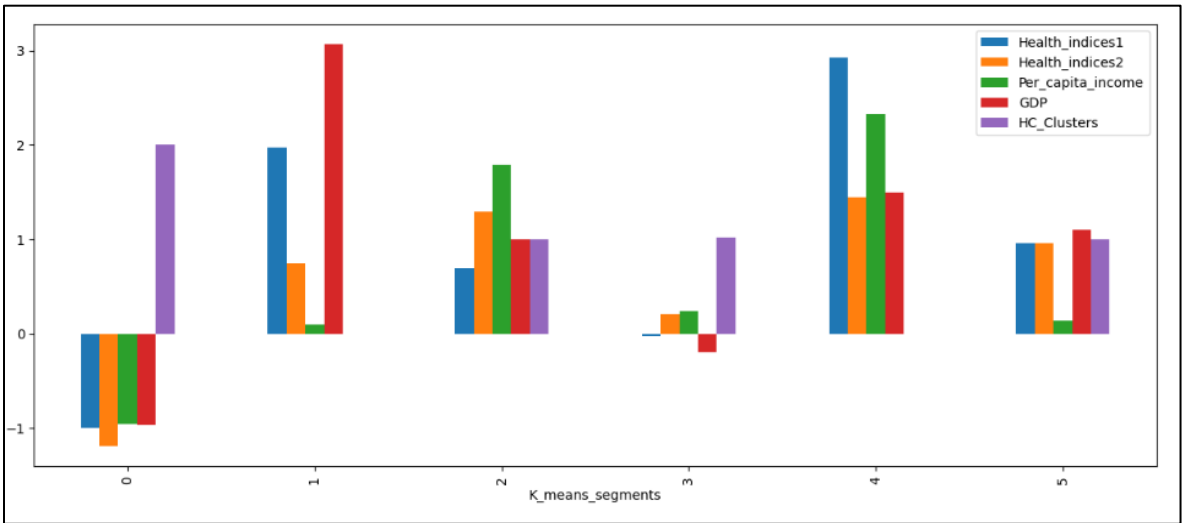
  Three clusters have been formed with hierarchical clustering, along with count in each segment.

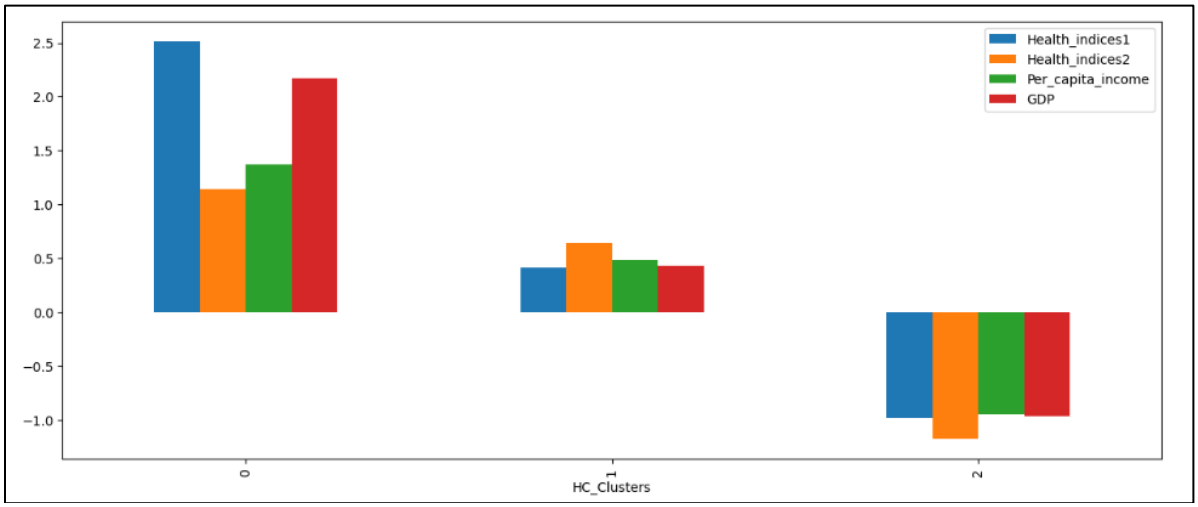
```
HC_segments
0                          4205.785714
1                          2880.890173
2                           751.981651
count_in_each_segment    [14, 173, 109]
Name: Per_capita_income, dtype: object
```

Topic 5: K-means vs. Hierarchical Clustering - K-means clustering and hierarchical clustering are two distinct approaches to grouping data points into clusters, and each has its own strengths and weaknesses. Here's a comparison between both:

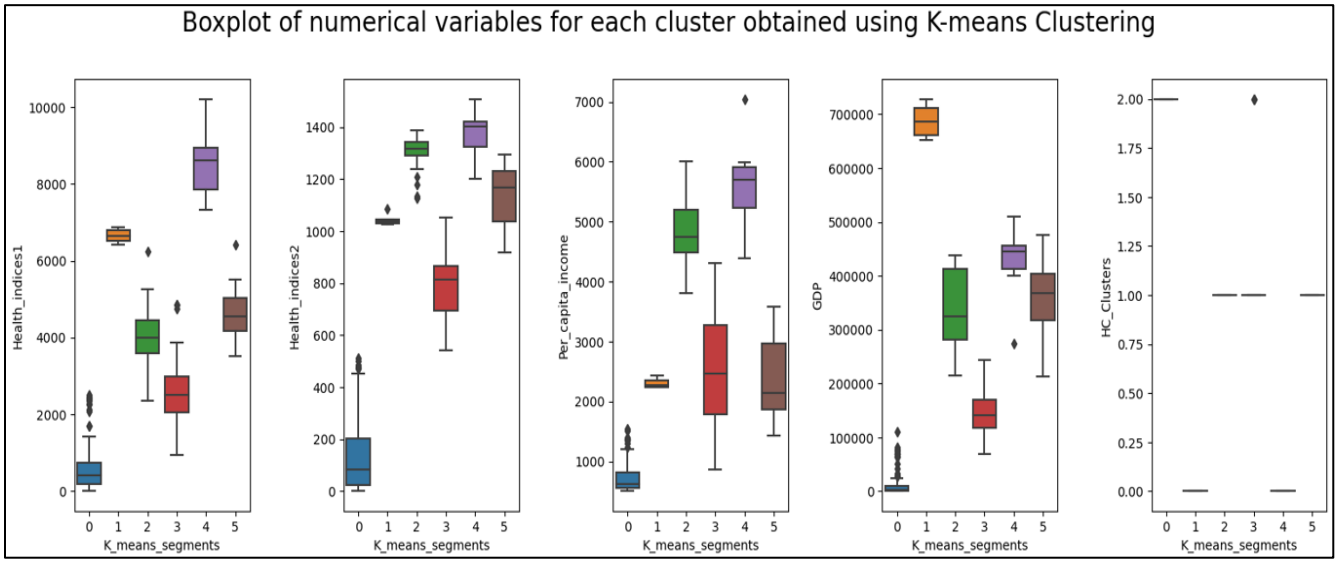
As per K_means the 5th cluster (index 4) has maximum values except for GDP and as per HC clustering cluster 1(index 0) stands the first with highest count in the segment.

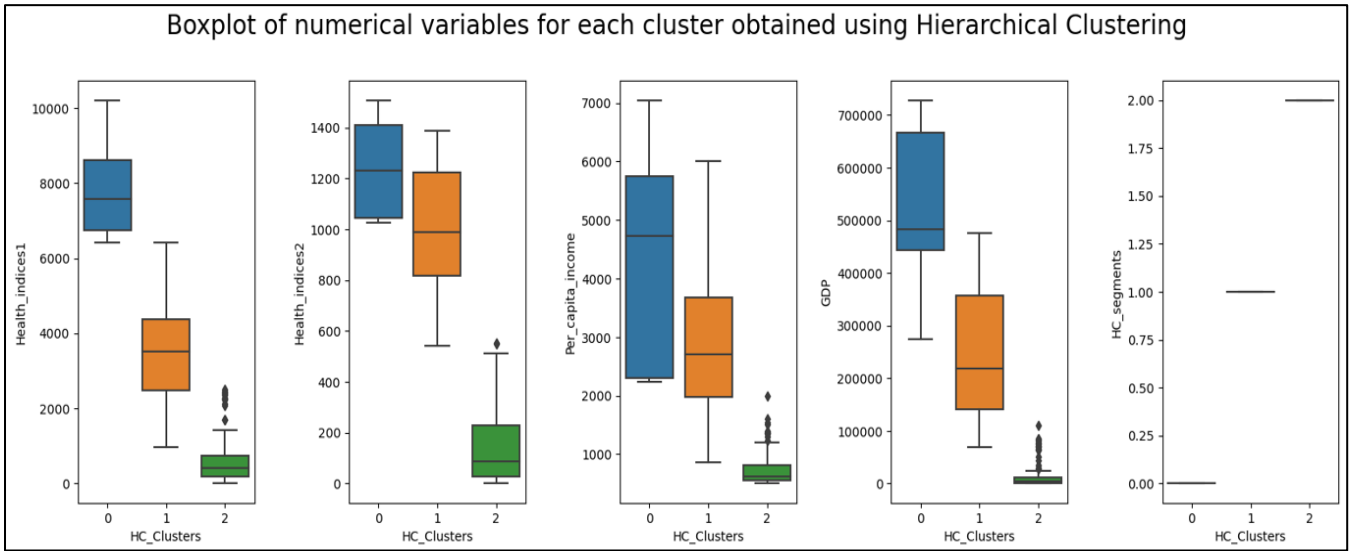
K_means_segments	Health_indices1	Health_indices2	Per_capita_income	GDP	HC_Clusters	count_in_each_segment
0	604.439252	137.000000	732.336449	12979.140187	2.000000	107
1	6649.333333	1044.000000	2299.833333	687649.666667	0.000000	6
2	4041.266667	1302.300000	4823.933333	341640.233333	1.000000	30
3	2577.444444	792.477778	2521.766667	142134.088889	1.022222	90
4	8589.375000	1371.000000	5635.250000	425379.625000	0.000000	8
5	4577.236364	1145.836364	2369.509091	359223.072727	1.000000	55





- **Let's create some plots on the original data to understand the customer distribution among the clusters.**





Topic 6: Actionable Insights and Recommendations:

- **Cluster Comparison**: From both the K-means & hierarchical clustering we get 3 as the optimal number of clusters.

- **Observations**: Three cluster group formed are: Cluster 1 – High GDP per capita area, Cluster 2 – Low GDP, Cluster 3 – Medium GDP.

- **Insights:**

    1. Cluster 1 – High GDP per capita area – Per capita income in these areas is very high, economic and health conditions are excellent here, and these areas have highest growth rate.

    2. Cluster 2 – Low GDP per capita area – Per capita income in these areas is very low, we notice a low growth rate, and economic & health condition in the area are not good.

    3. Cluster 3 – Medium GDP per capita area – Per capita income is average. We observe a moderate growth rate, and health & economic condition is satisfactory in the area.

- **Business Recommendations:** The important features that affect Health & Economic conditions are workforce and productivity, higher these attribute higher the GDP, and thus better the economic and health conditions.

    1. Cluster 1 – High GDP per capita area – The size of workforce and maintaining growth in productivity will keep both economic and health conditions high.

    2. Cluster 2 – Low GDP – To increase size of workforce more employment opportunities need to be created which will increase productivity eventually.

    3. Cluster 3 – Medium GDP per capita area – New businesses in the area will help with growth and development in the area.

Part 2: PCA

Problem Statement

Context

The 'Hair Salon.csv' dataset contains various variables used for the context of Market Segmentation. This particular case study is based on various parameters of a salon chain of hair products. You are expected to do Principal Component Analysis for this case study according to the instructions given in the rubric.

Objective

Apply Principal Component Analysis (PCA) on the 'Hair Salon.csv' dataset, which encompasses various variables related to a salon chain's market segmentation. The goal is to analyze and interpret the principal components

Data Description

1. ProdQual: Product Quality
2. Ecom: E-Commerce
3. TechSup: Technical Support
4. CompRes: Complaint Resolution
5. Advertising: Advertising
6. ProdLine: Product Line
7. SalesFImage: Salesforce Image
8. ComPricing: Competitive Pricing
9. WartyClaim: Warranty & Claims
10. OrdBilling: Order & Billing
11. DelSpeed: Delivery Speed
12. Satisfaction: Customer Satisfaction

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Data Analysis & A detailed solution around the problems:

Observation 1: The dataset has 100 rows and 13 columns, we drop 2 columns (ID & Satisfaction) that won't be part of the data analyzing.

Observation 2: All the columns in the data are non-null. We have data in float-type only which will be easy for analyzing the data further.

Observation 3: It's very crucial to identify the missing values, and treat them before moving ahead with the data analysis, and as checked there are **no missing values** in the data, and we are good to proceed.

Observation 4: The unique data in each column is as follows:

```
ProdQual        43
Ecom            27
TechSup         50
CompRes         45
Advertising     41
ProdLine        42
SalesFImage     35
ComPricing      45
WartyClaim      34
OrdBilling      37
DelSpeed        30
```

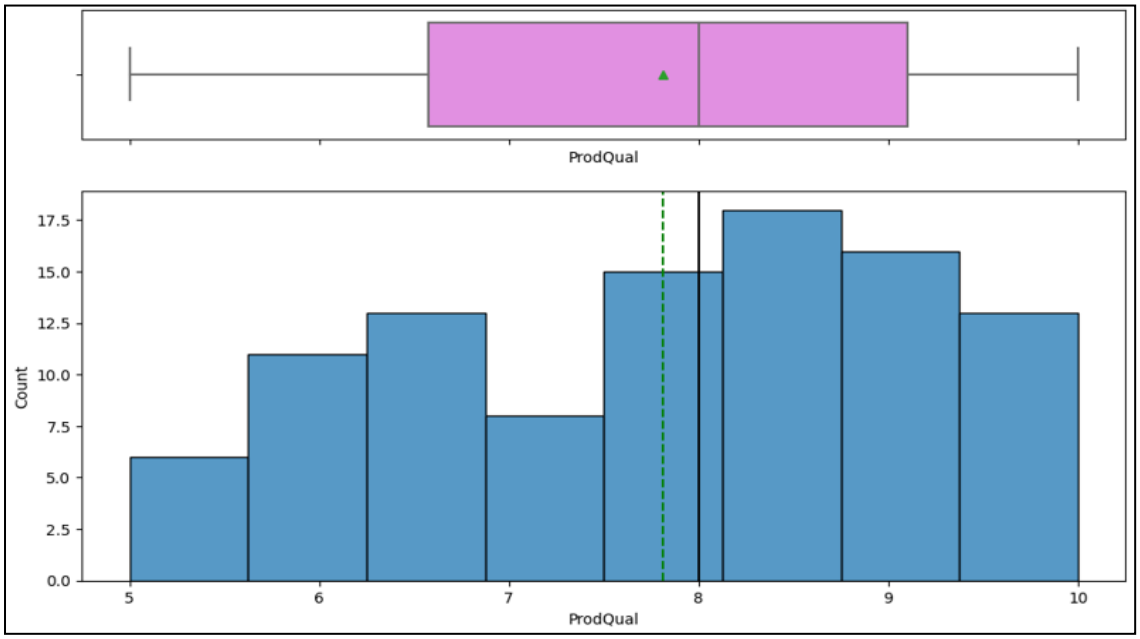
Observation 5: Statistical summary is nothing but the technique to identify variables such as min, max, mean, standard deviation, count, 25th, 50th, & 75th percentile of the data.

	count	mean	std	min	25%	50%	75%	max
ProdQual	100.0	7.810	1.396279	5.0	6.575	8.00	9.100	10.0
Ecom	100.0	3.672	0.700516	2.2	3.275	3.60	3.925	5.7
TechSup	100.0	5.365	1.530457	1.3	4.250	5.40	6.625	8.5
CompRes	100.0	5.442	1.208403	2.6	4.600	5.45	6.325	7.8
Advertising	100.0	4.010	1.126943	1.9	3.175	4.00	4.800	6.5
ProdLine	100.0	5.805	1.315285	2.3	4.700	5.75	6.800	8.4
SalesFImage	100.0	5.123	1.072320	2.9	4.500	4.90	5.800	8.2
ComPricing	100.0	6.974	1.545055	3.7	5.875	7.10	8.400	9.9
WartyClaim	100.0	6.043	0.819738	4.1	5.400	6.10	6.600	8.1
OrdBilling	100.0	4.278	0.928840	2.0	3.700	4.40	4.800	6.7
DelSpeed	100.0	3.886	0.734437	1.6	3.400	3.90	4.425	5.5

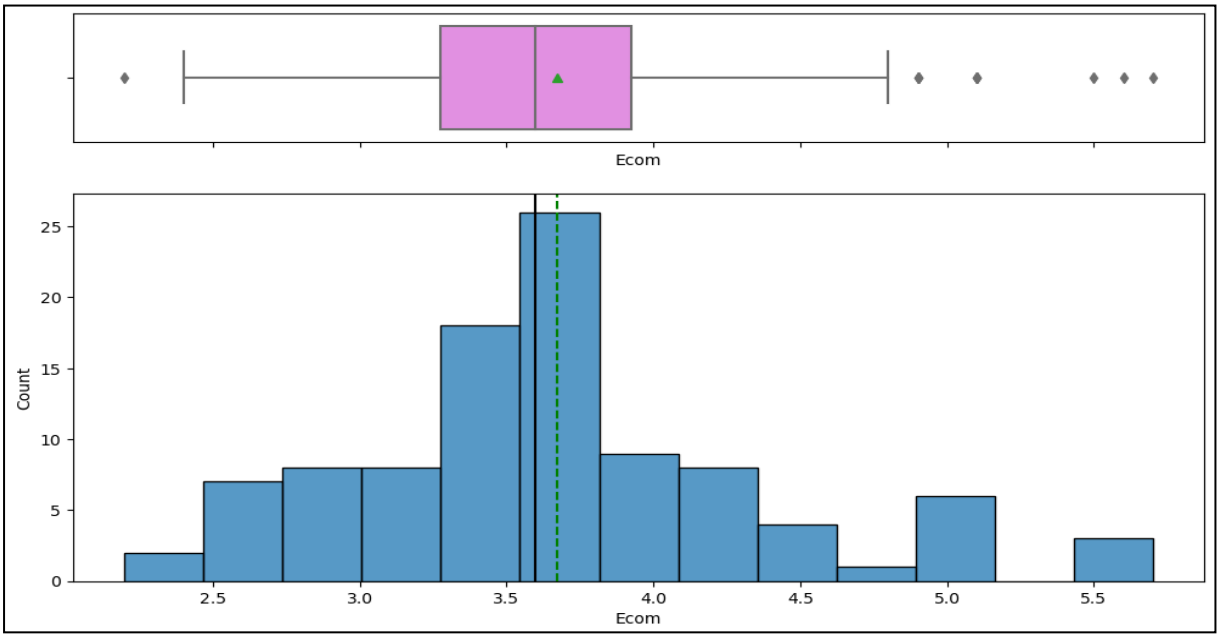
Exploratory Data Analysis

1. Univariate Analysis: The green dotted line denotes the average of the data.

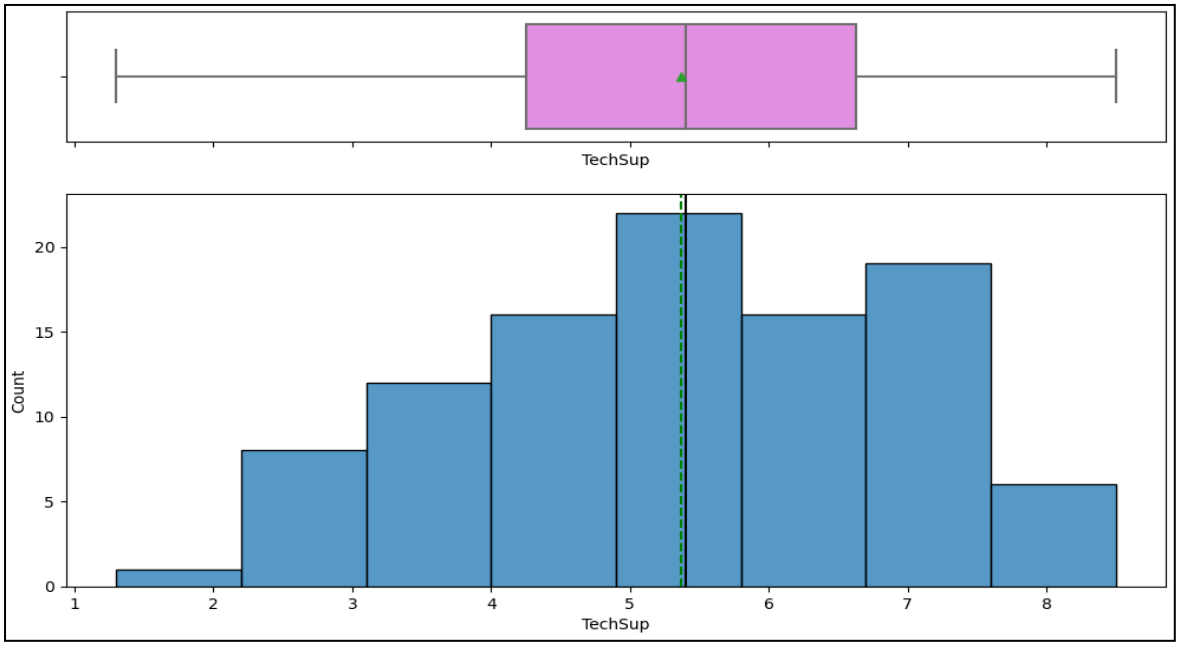
1.1 Product Quality: The data has no outliers and is almost evenly spread from 5 to 10.



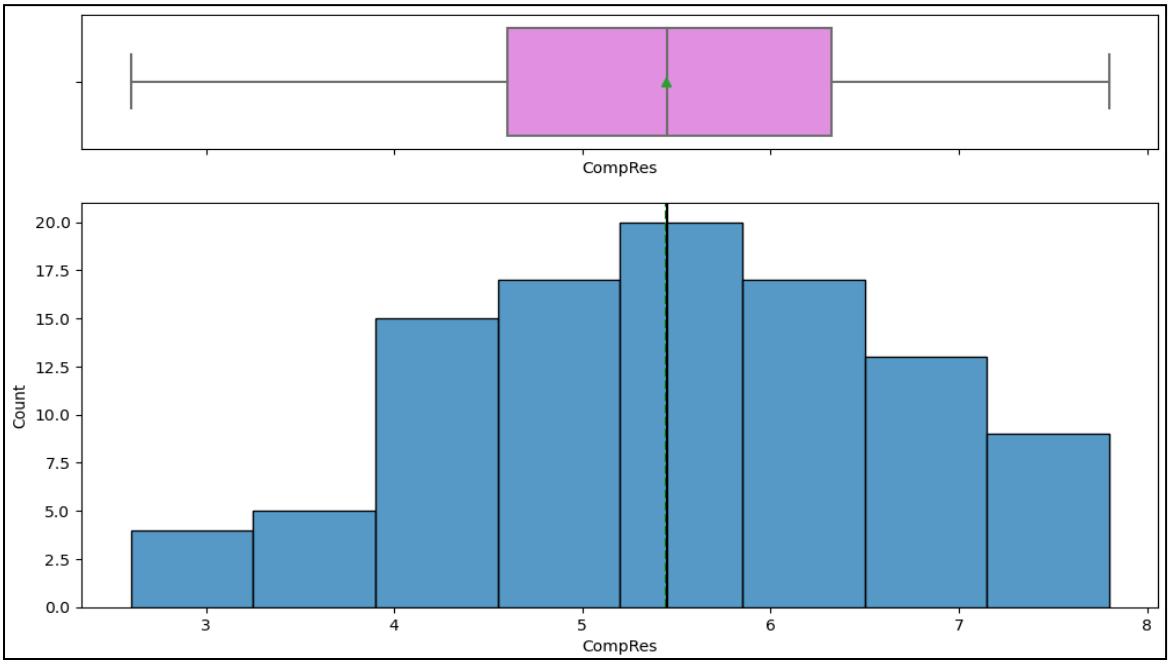
1.2 E-commerce: The data has outliers and 60% of the data ranges from 3.3 to 3.9.



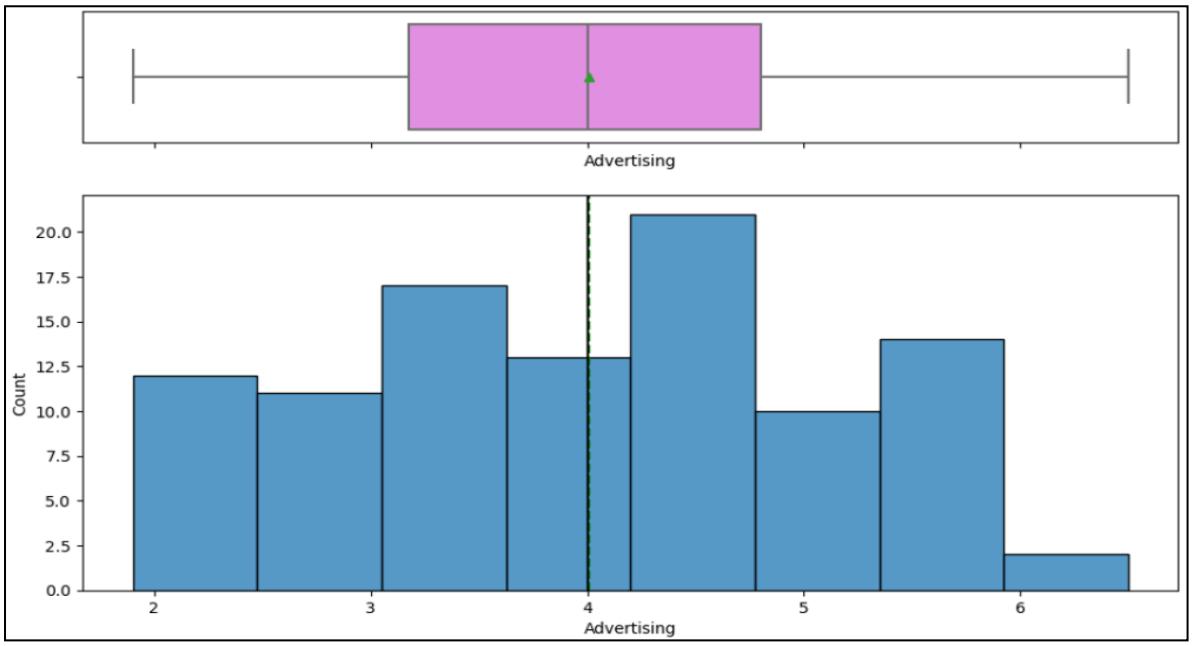
1.3 Technical Support: No outliers found in the tech support, data quite spread throughout, with 74% data ranging from 4 to 7.6.



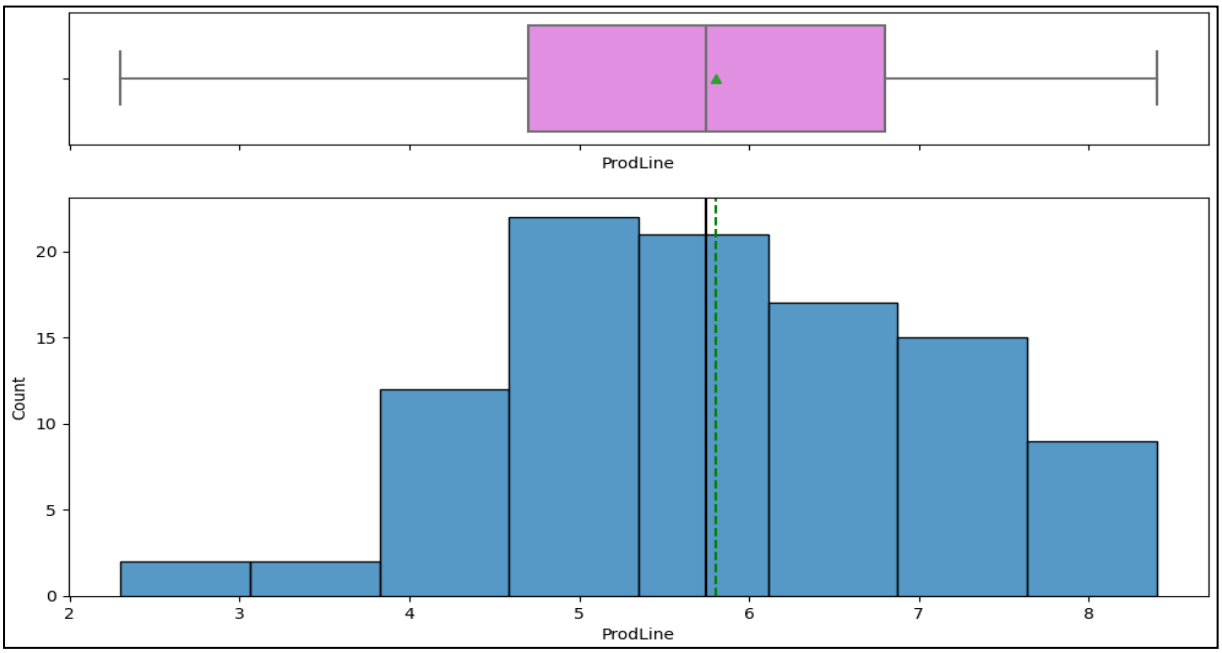
1.4 Complaint Resolution: No outliers found, data quite spread throughout with maximum ranging between 3.9 to 7.1 (82%).



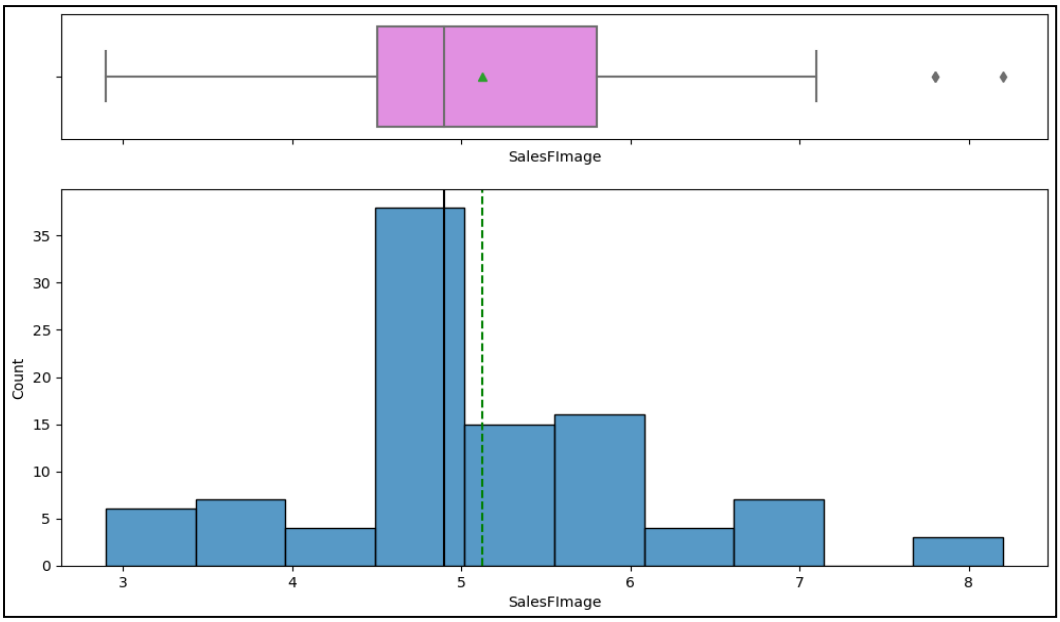
1.5 Advertising: No outliers found, data quite consistent from 1.9 to 5.9 which is 98% of the data.



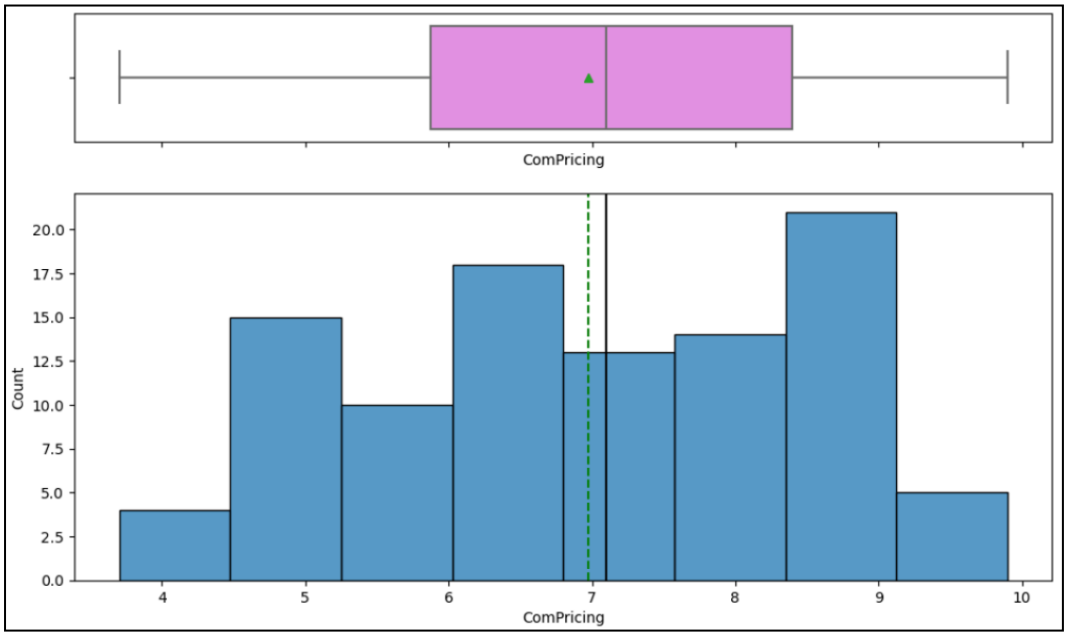
1.6 Product Line: No outliers found, data accumulating more from 3.9 onwards to 7.6 forming 87% of the data.



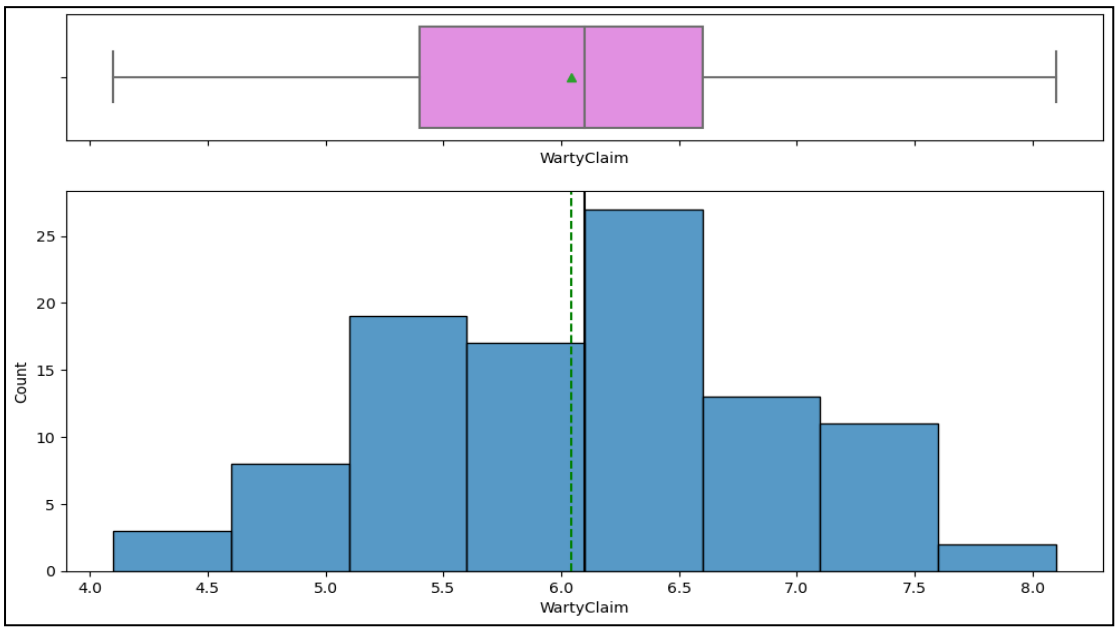
1.7 Sales force Image: Few outliers found in the data, data accumulates the maximum between 4.5 to 5, and others spread throughout.



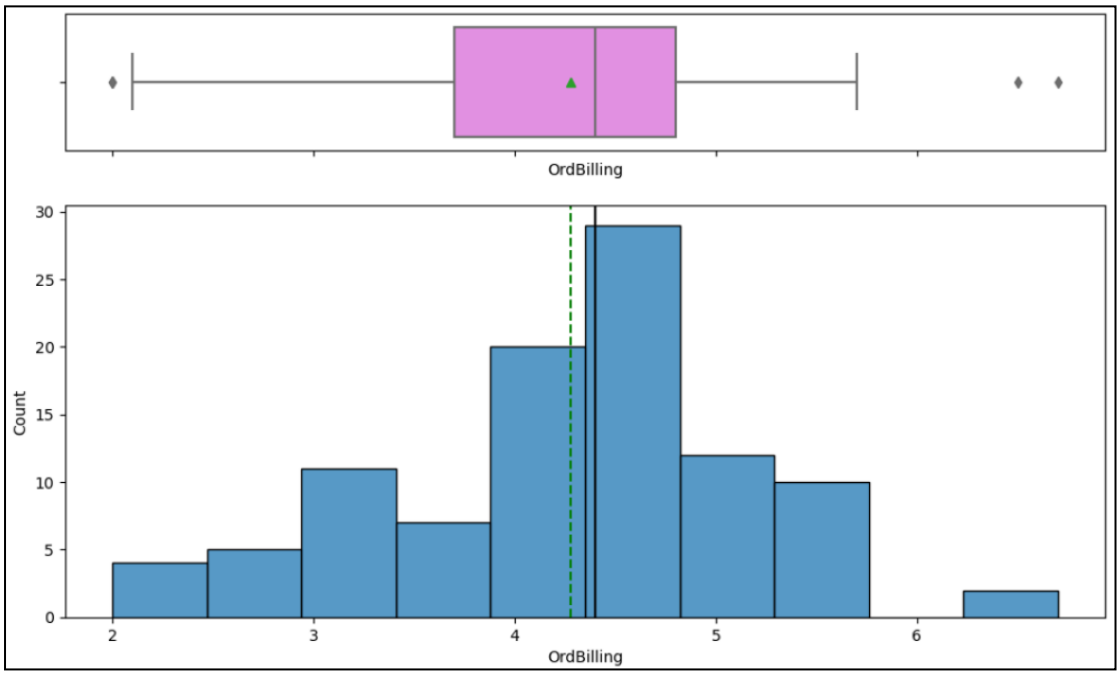
1.8 Competitive Pricing: No outliers found in the data below, data quite evenly spread from 4.5 to 9.1



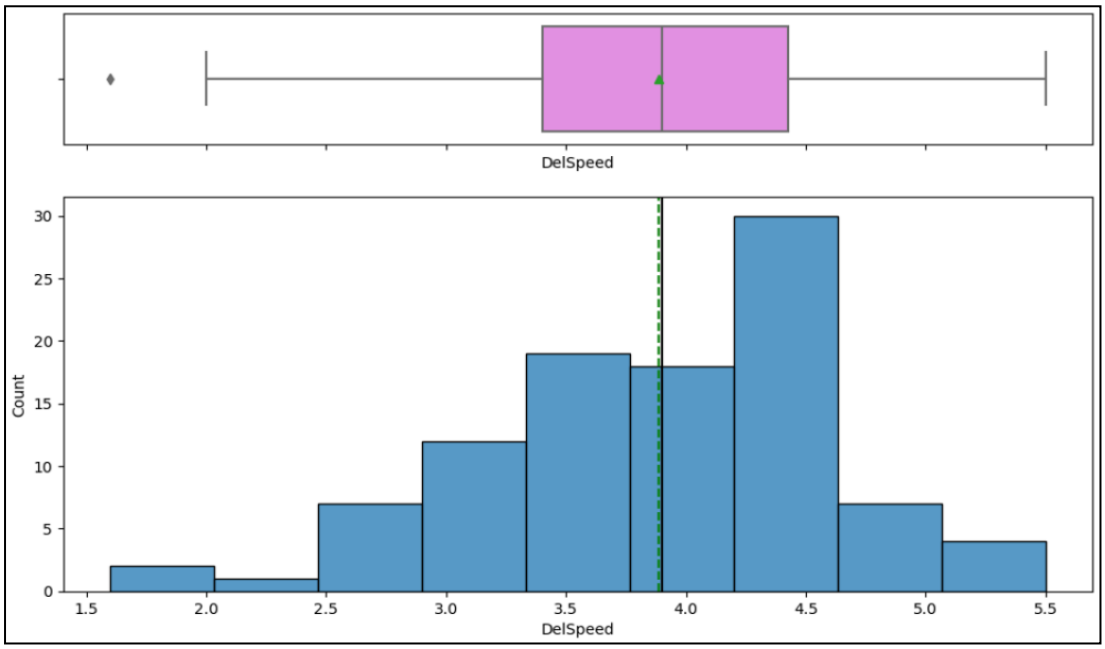
1.9 Warranty & Claims: No outliers find here, data ranging the highest bet. 5.1 to 7.6



1.10 Order & Billing: Some outliers found in the order and billing. The histogram bars are high at 3.9 to 4.8 with total of 49 count forming 49% of the data.



1.11 Delivery Speed: No outliers found in the data of delivery speed given below.



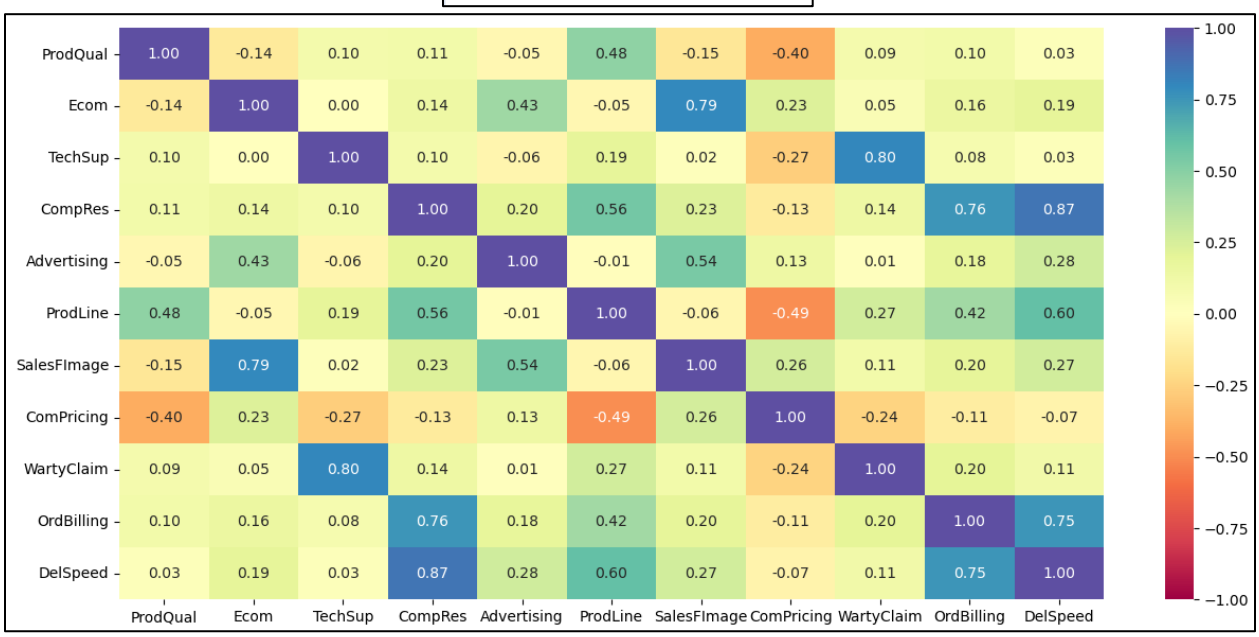
2. <u>Bivariate Analysis:</u> Pairplot: We see very less correlation between the categories, it's either negative or least positively related.

	ProdQual	Ecom	TechSup	CompRes	Advertising	ProdLine	SalesFImage	ComPricing	WartyClaim	OrdBilling	DelSpeed
ProdQual	1.000000	-0.137163	0.095600	0.106370	-0.053473	0.477493	-0.151813	-0.401282	0.088312	0.104303	0.027718
Ecom	-0.137163	1.000000	0.000867	0.140179	0.429891	-0.052688	0.791544	0.229462	0.051898	0.156147	0.191636
TechSup	0.095600	0.000867	1.000000	0.096657	-0.062870	0.192625	0.016991	-0.270787	0.797168	0.080102	0.025441
CompRes	0.106370	0.140179	0.096657	1.000000	0.196917	0.561417	0.229752	-0.127954	0.140408	0.756869	0.865092
Advertising	-0.053473	0.429891	-0.062870	0.196917	1.000000	-0.011551	0.542204	0.134217	0.010792	0.184236	0.275863
ProdLine	0.477493	-0.052688	0.192625	0.561417	-0.011551	1.000000	-0.061316	-0.494948	0.273078	0.424408	0.601850
SalesFImage	-0.151813	0.791544	0.016991	0.229752	0.542204	-0.061316	1.000000	0.264597	0.107455	0.195127	0.271551
ComPricing	-0.401282	0.229462	-0.270787	-0.127954	0.134217	-0.494948	0.264597	1.000000	-0.244986	-0.114567	-0.072872
WartyClaim	0.088312	0.051898	0.797168	0.140408	0.010792	0.273078	0.107455	-0.244986	1.000000	0.197065	0.109395
OrdBilling	0.104303	0.156147	0.080102	0.756869	0.184236	0.424408	0.195127	-0.114567	0.197065	1.000000	0.751003
DelSpeed	0.027718	0.191636	0.025441	0.865092	0.275863	0.601850	0.271551	-0.072872	0.109395	0.751003	1.000000

Heatmap: Now these correlations give a clear picture on what variables are highly correlated with each other.

- Starting from delivery speed and complaint resolution, which means they are related and affect each other.
- Second technical support and warranty & claims are quite correlated with an equal amount of correlation as e-commerce and salesforce image.

		correlation
DelSpeed	CompRes	0.865092
TechSup	WartyClaim	0.797168
Ecom	SalesFImage	0.791544
OrdBilling	CompRes	0.756869
DelSpeed	OrdBilling	0.751003
DelSpeed	ProdLine	0.601850
CompRes	ProdLine	0.561417
SalesFImage	Advertising	0.542204
ComPricing	ProdLine	0.494948
ProdQual	ProdLine	0.477493
Advertising	Ecom	0.429891
ProdLine	OrdBilling	0.424408
ProdQual	ComPricing	0.401282



3. PCA Transformation:

3.1 Scaled data is as below: Scaling is an essential step when applying Principal Component Analysis (PCA) because it ensures that all features contribute equally to the analysis. Since PCA is based on the variance-covariance matrix, features with larger scales may dominate the analysis and lead to misleading results.

	ProdQual	Ecom	TechSup	CompRes	Advertising	ProdLine	SalesFImage	ComPricing	WartyClaim	OrdBilling	DelSpeed
0	0.496660	0.327114	-1.881421	0.380922	0.704543	-0.691530	0.821973	-0.113185	-1.646582	0.781230	-0.254531
1	0.280721	-1.394538	-0.174023	1.462141	-0.544014	1.600835	-1.896068	-1.088915	-0.665744	-0.409009	1.387605
2	1.000518	-0.390241	0.154322	0.131410	1.239639	1.218774	0.634522	-1.609304	0.192489	1.214044	0.840226
3	-1.014914	-0.533712	1.073690	-1.448834	0.615361	-0.844354	-0.583910	1.187789	1.173327	0.023805	-1.212443
4	0.856559	-0.390241	-0.108354	-0.700298	-1.614207	0.149004	-0.583910	-0.113185	0.069885	0.240212	-0.528220

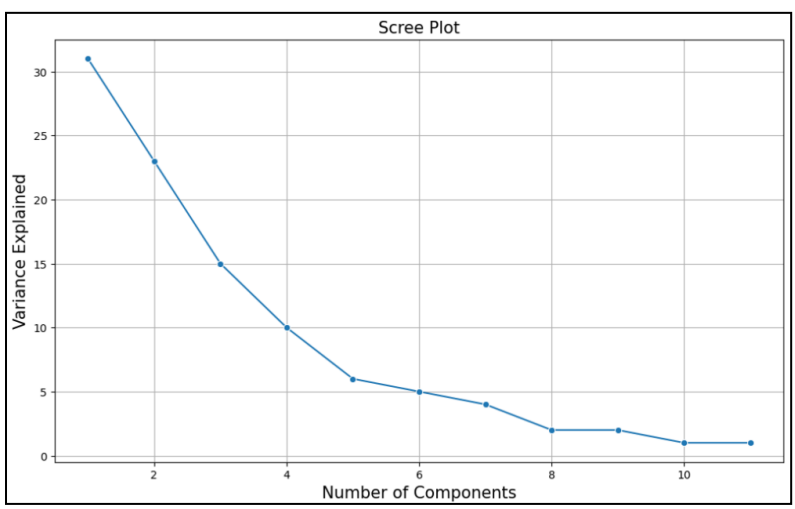
3.2 Checking the outliers after scaling:

- **Step 1**: Create the covariance Matrix

	ProdQual	Ecom	TechSup	CompRes	Advertising	ProdLine	SalesFImage	ComPricing	WartyClaim	OrdBilling	DelSpeed
ProdQual	1.01	-0.14	0.10	0.11	-0.05	0.48	-0.15	-0.41	0.09	0.11	0.03
Ecom	-0.14	1.01	0.00	0.14	0.43	-0.05	0.80	0.23	0.05	0.16	0.19
TechSup	0.10	0.00	1.01	0.10	-0.06	0.19	0.02	-0.27	0.81	0.08	0.03
CompRes	0.11	0.14	0.10	1.01	0.20	0.57	0.23	-0.13	0.14	0.76	0.87
Advertising	-0.05	0.43	-0.06	0.20	1.01	-0.01	0.55	0.14	0.01	0.19	0.28
ProdLine	0.48	-0.05	0.19	0.57	-0.01	1.01	-0.06	-0.50	0.28	0.43	0.61
SalesFImage	-0.15	0.80	0.02	0.23	0.55	-0.06	1.01	0.27	0.11	0.20	0.27
ComPricing	-0.41	0.23	-0.27	-0.13	0.14	-0.50	0.27	1.01	-0.25	-0.12	-0.07
WartyClaim	0.09	0.05	0.81	0.14	0.01	0.28	0.11	-0.25	1.01	0.20	0.11
OrdBilling	0.11	0.16	0.08	0.76	0.19	0.43	0.20	-0.12	0.20	1.01	0.76
DelSpeed	0.03	0.19	0.03	0.87	0.28	0.61	0.27	-0.07	0.11	0.76	1.01

- **Step 2**: Get eigen values and eigen vector: **Eigenvalues: [3.46 2.58 1.71 1.1 0.62 0.56 0.41 0.25 0.21 0.13 0.1]**

- **Step 3**: View Scree Plot to identify the number of components to be built



- **Step 4**: Apply PCA for the number of decided components to get the loadings and component output.

	ProdQual	Ecom	TechSup	CompRes	Advertising	ProdLine	SalesFImage	ComPricing	WartyClaim	OrdBilling	DelSpeed
PC0	-0.130000	-0.170000	-0.160000	-0.470000	-0.180000	-0.390000	-0.200000	0.150000	-0.210000	-0.440000	-0.470000
PC1	-0.310000	0.450000	-0.230000	0.020000	0.360000	-0.280000	0.470000	0.410000	-0.190000	0.030000	0.070000

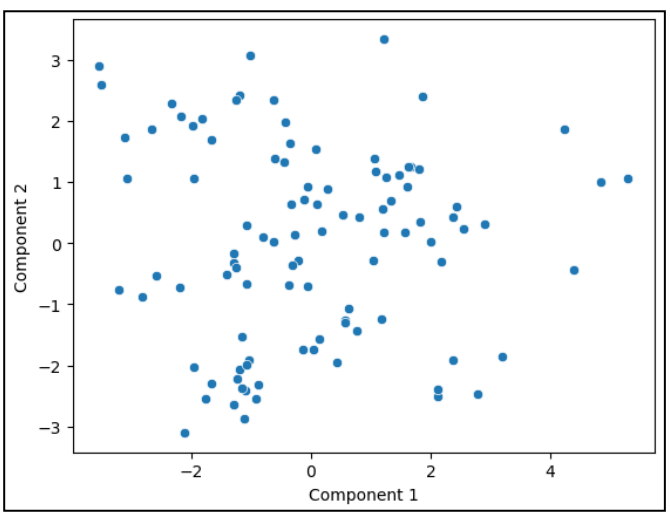
4. <u>Interpretation of Principal Components & Variance Explanation:</u> Let's see how each principal component contributes to the variation in the data.

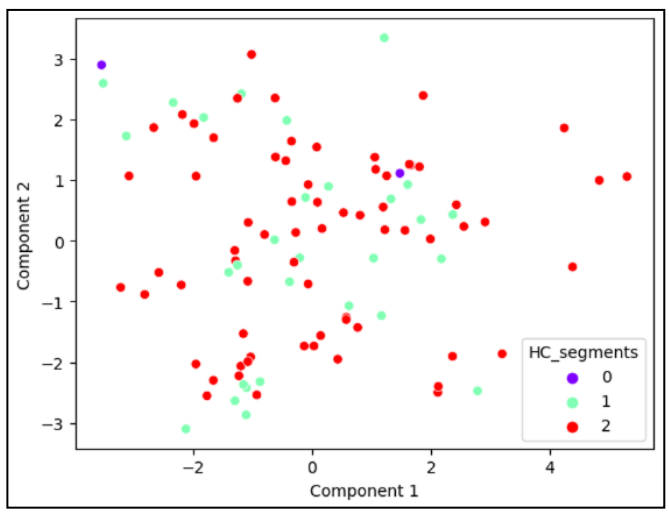
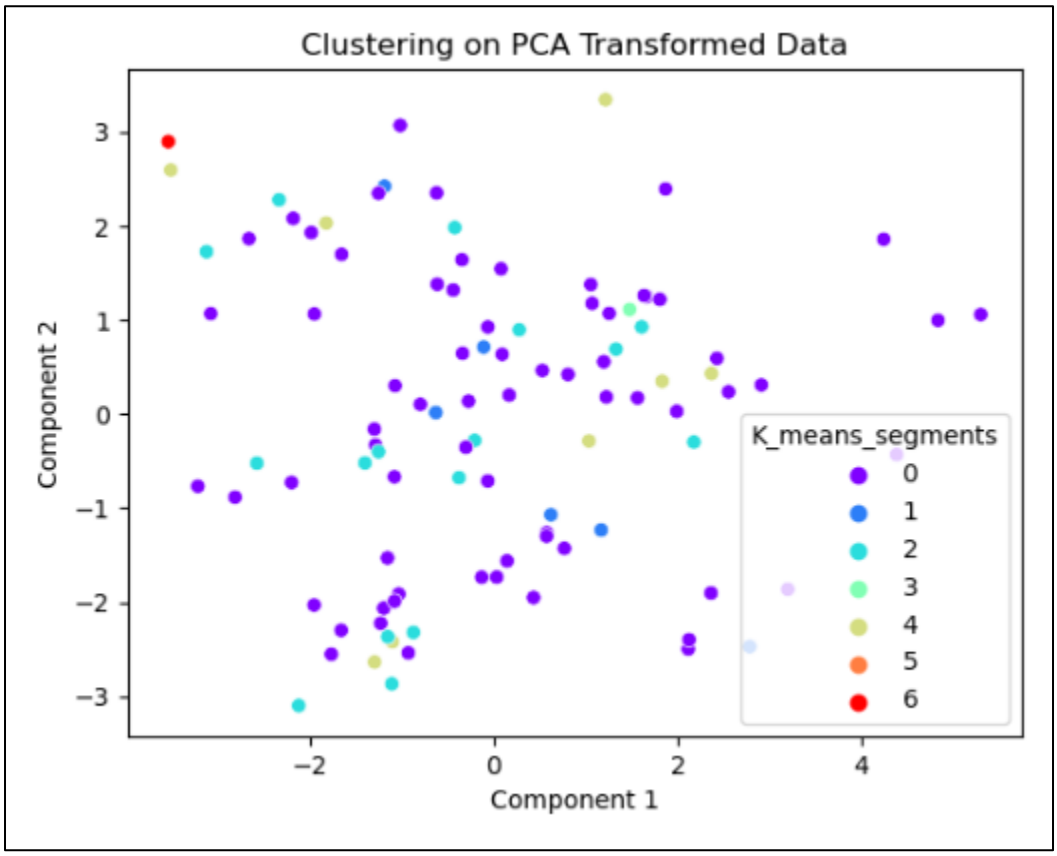
Observation: The first two principal components explain 54.34% of the variance in the data.

- **Explained Variance Ratio:** [0.31154285 0.2318997]
- **Principal Component 1 Loadings**: [-0.13378962 -0.16595278 -0.15769263 -0.47068359 -0.18373495 -0.38676517 -0.2036696 0.15168864 -0.21293363 -0.43721774 -0.47308914]
- **Principal Component 2 Loadings**: [-0.31349802 0.44650918 -0.23096734 0.01944394 0.36366471 -0.28478056 0.47069599 0.4134565 -0.19167191 0.02639905 0.07305172]
- **Cumulative Explained Variance**: [0.31154285 0.54344255]

5.<u>Visualsiation:</u>

- We observe three clusters formed with K-means clustering on both original & PCA transformed data.





6. Dimensionality Reduction Impact: The impact of PCA-based dimensionality reduction on the performance of the clustering algorithm used earlier gives clustering results after applying PCA and shows improvements in the identified customer segments.